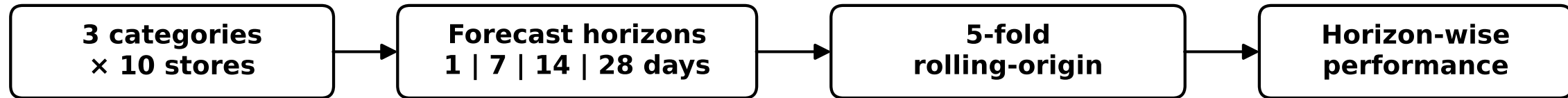
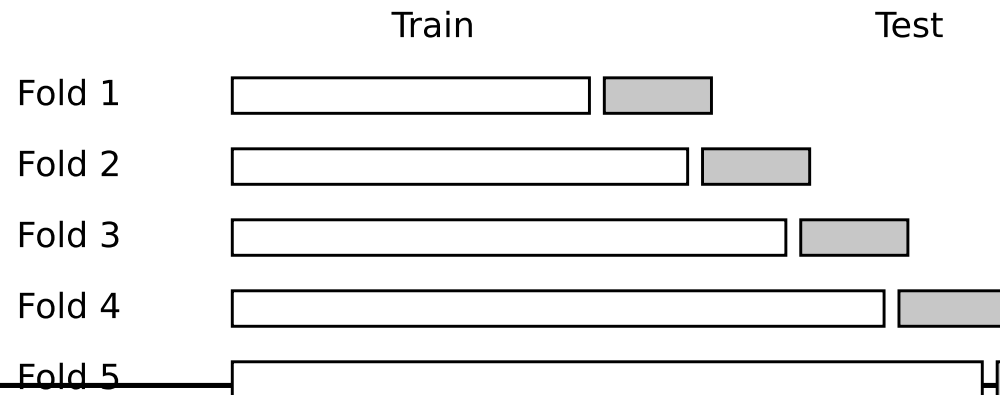


Two-Layer Experimental and Rolling-Origin Validation Design

A. Category-store horizon-aware experiment

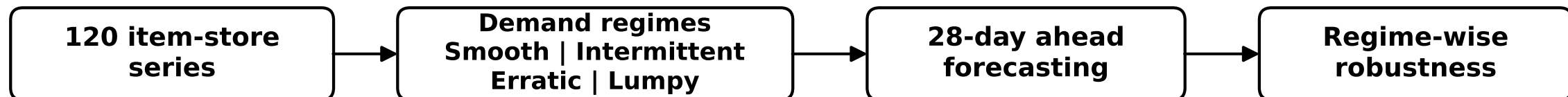


Expanding-window validation timeline



Both layers preserve temporal order; random train-test splitting is avoided.

B. Item-store regime-sensitive experiment



Expanding-window validation timeline

